

C5 - Scraping off leftover asphalt mix/fines from the scoop into sample container.



C6 - Scrapping off leftover asphalt mix/fines from the template into sample container.



C7 - For a bigger sampling template, a shovel can be used.

APPENDIX 502-C

BLANK AGGREGATE SAMPLE PREPARATION

502-C.01 Blank aggregate samples replicate the Job Mix Formula design aggregate gradation, without asphalt cement and are used to correlate the Ministry, Contractor, and appeal lab ignition ovens.

502-C.02 Twelve (12) blanks are required, each weighing 1900 ± gram. This size approximates the size of the mix samples, less the asphalt cement, that will be used for the project's daily Quality Assurance testing.

502-C.03 The blanks shall be prepared from bulk aggregate samples (i.e. coarse aggregates, fine aggregates, blend sand, etc.) that were produced from the same sources and are the same materials as the aggregates that will be used for the project and have been designated in the Mix Design. The amount of each aggregate material required will be 50 kg to 60 kg.

502-C.04 The bulk aggregate samples shall first be dried, and then thoroughly blended at the design proportions. The blended material shall then be sieved into each of the individual sieve sizes designated in the Mix Design down to passing 0.075 mm.

502-C.05 The aggregate blanks will then be made up from the sieved material, starting with the pan, and progressively adding proportions from each successive individual sieve to conform to the estimated dry aggregate JMF gradation within the tolerances specified below.

(a) Prepare two (2) additional dry blank samples within the tolerances specified in Table 502-R. Run washed sieve analysis on both samples and using the average passing on the two samples to arrive at a "Target Adjustment", matching the submitted design JMF for the final blank aggregate preparation.

(b) Masses of aggregate added shall be done based on the average washed sieve adjustment as described above.

(c) Masses of aggregate added shall be within the tolerances specified in Table 502-R.

502-C.06 Samples shall be sealed in cans or bagged in moisture-proof heavy gauge plastic bags, and labeled with the project number, blank number, the preparation date, and a JMF identification number.

Table 502-R: APPENDIX 502-C - Cumulative Mass Tolerance by Sieve Size

Sieve Size	Cumulative Mass Tolerance (g)
Pan	0.1
0.075	0.1
0.15	0.1
0.30	0.1
0.60	0.1
1.18	0.1
2.36	0.1
4.75	0.1
9.50	0.7
12.5	2.0
16.0	3.0
19.0	6.0
25.0	13.0
37.5	45.0

APPENDIX 502-D

IGNITION OVEN CORRELATION PROCEDURE

502-D.01 Asphalt Mix samples are prepared and burned to determine the effect each oven has on the mix.

502-D.02 The Contractor and the Ministry shall each prepare three (3) Asphalt Mix calibration samples by adding AC (provided by the Contractor from the supplier and batch that will be used on the project) to randomly selected aggregate blanks prepared by the Contractor in accordance with APPENDIX 502-C.

502-D.03 AC shall be added to the blanks at the proportion specified in the accepted JMF, within a tolerance of ± 0.1 gram.

502-D.04 Each party shall burn their mix calibration samples in their ignition oven in accordance with ASTM D6307 to determine their calibration factor.

502-D.05 The calibration factors determined above shall be compared and the inter-laboratory correlation determined.

502-D.06 Correlation between the Contractor's lab and the Ministry lab shall be completed, reviewed and mutually acceptable to the Contractor and the Ministry Representative a minimum of one (1) working day prior to placement of any mix.

502-D.07 In the event of an appeal on AC Content, the appeal laboratory shall also prepare and burn calibration samples to determine their oven calibration factor.

502-D.08 Results from testing of any asphalt mix shall report the measured AC Content and the corrected AC Content after applying the laboratory calibration factor.

APPENDIX 502-E

ASPHALT FILM THICKNESS IN BITUMINOUS MIXTURES

502-E.01 The following method is to be used to determine the asphalt film thickness on aggregate particles in an asphalt mix. The results of this calculation express the asphalt film thickness in μm (microns).

502-E.02 Data Required:

(a) Mix Design percentage of asphalt cement by weight of dry aggregate

(b) Percentage of Asphalt Absorption

(c) Specific gravity of aggregate

(d) Gradation of asphalt mix aggregate

(e) Specific Gravity of the asphalt cement

Table 502-S: APPENDIX 502-E - Surface Area (SA) Factors

Total % Passing Sieve No.	Maximum Size	4.75 mm (No. 4)	2.36 mm (No. 8)	1.18 mm (No. 16)	0.600 mm (No. 30)	0.300 mm (No. 50)	0.150 mm (No. 100)	0.075 mm (No. 200)
S.A. (m^2/kg)	0.41	0.41	0.82	1.64	2.87	6.14	12.29	32.77
Surface Area factors shown are applicable only when all the above listed sieves are used in the sieve analysis.								

This example tabulation demonstrates the calculation of surface area by this method.

Table 502-T: APPENDIX 502-E - Example Surface Area Calculation

Sieve Size (mm)	Percent Passing	Surface Area Factor (m^2/kg)	Surface Area (m^2/kg)
19.0	*100*	0.41	0.41
12.5	*95*		
9.5	*80*		
4.75	*57*	0.41	*0.23*
2.36	*39*	0.82	*0.32*
1.18	*31*	1.64	*0.51*
0.600	*23*	2.87	*0.66*
0.300	*17*	6.14	*1.04*
0.150	*12*	12.29	*1.47*
0.075	*7.5*	32.77	*2.46*
Surface Area of Sample (sum of the above):			***7.1***

Equation 502-4: $$Corrected\ Surface\ Area = \frac{2.650}{Specific\ Gravity\ of\ the\ Aggregate} \text{ x } Surface\ Area\ of\ Sample$$

Equation 502-5: $$Effective\ Asphalt\ Cement\ Content = Mix\ Design\ AC - Asphalt\ Absorption$$

Equation 503-6: $$Film\ Thickness = 10 \text{ x } \frac{Effective\ Asphalt\ Cement\ Content}{Corrected\ Surface\ Area * Specific\ Gravity\ of\ Asphalt\ Cement}$$

APPENDIX 502-F

MARSHALL HAMMER CORRELATION PROCEDURE

502-F.01 Overview – This procedure is used to correlate the mechanical Marshall hammers of both the Ministry and the Contractor, to produce density results that would be representative of testing on the Ministry's manual Marshall hammer.

For a given mix, this procedure gives the following results:

(a) **DManual =** the average density obtained from the Ministry's manual Marshall hammer at the nominal number of blows.

(b) **NMin and NCon =** the respective number of blows to be delivered by the Ministry's and Contractor's mechanical Marshall hammers.

(c) **CMin and CCon =** the respective density correlations for the Ministry and Contractor's mechanical Marshall hammers, to be added to subsequent density test results to approximate the densities that would be expected had the Ministry's manual Marshall hammer been used.

502-F.02 General:

(a) Any Marshall hammer must be correlated for each mix used on each project and every time the hammer is relocated.

(b) The Marshall hammer correlations shall be determined prior to the production of any mix.

(c) Briquettes shall be prepared and tested in accordance with ASTM D6926.

(d) All densities shall be determined in in accordance with ASTM D2726.

(e) The Nominal Number of Blows ("NNom") to be applied using the Ministry's manual hammer shall be as appropriate to the pavement design requirement (Typically 75 blows for heavy traffic, although either 35 or 50 blows may be used for lighter traffic).

502-F.03 Mix Sample Preparation:

(a) The Contractor shall prepare a minimum of thirty 1200 g (30 x 1200 g) Asphalt Mix Aggregate blanks using the APPENDIX 502-C procedures, supplying eighteen (18) to the Ministry and retaining twelve (12) for the its own use.

(b) The Contractor shall supply a minimum of 2 kg of the specified AC to the Ministry and retain a similar amount for its own use.

(c) AC shall be added to the aggregate blanks at the trial JMF rate (±1 gram); by the Ministry to each of fifteen (15) blanks and by the Contractor to each of twelve (12) blanks.

(d) Three (3) of the aggregate blanks shall be retained by the Ministry for substitute samples or additional samples, if needed.

502-F.04 Ministry Sample Testing

(a) Three (3) of the Ministry's mix samples shall be compacted at NNom blows using the Ministry's manual Marshall Hammer and the average density determined ("Dmanual").

(b) Twelve (12) of the mix samples shall be compacted using the mechanical Marshall hammer. Four (4) sets of three (3) samples each shall respectively receive:

(i) NNom minus 10 blows,

(ii) NNom blows,

(iii) NNom plus 10 blows,

(iv) NNom plus 20 blows.

(c) The above numbers of blows may be varied at the discretion of the operator to better bracket the anticipated blows required to achieve DManual.

(d) The average densities for each set of three (3) samples shall be calculated.

(e) The average densities calculated above will be plotted against the number of blows received, and the best-fit curve drawn through them. If using computer-assisted calculations, a second-degree polynomial curve fit is to be used.

(f) Where the curve established above is considered to be a poor fit, a fifth set of three (3) mix samples shall be prepared and compacted at a number of blows appropriate to best supplement the existing data. Typically, the number of blows used would be close to the number estimated to produce a density close to DManual. The results shall be added to the data set and a revised curve fitted.

(g) Using the plotted curve or its formula, the number of blows of the mechanical hammer required to achieve the average Ministry manual hammer density shall be determined.

(h) The above number shall be rounded to the nearest whole number to establish "NMin", the number of blows to be delivered by the Ministry's mechanical hammers in all subsequent testing.

502-F.05 Contractor Sample Testing

(a) The Contractor shall repeat steps Appendix 502-F.04(b) through Appendix 502-F.04(h) inclusive on twelve (12) additional mix samples using the Contractor's mechanical hammer and determine "NCon", the number of blows to be delivered by the Contractor's mechanical hammers in all subsequent testing.

(b) If considered necessary, the Contractor may prepare and test a fifth set of samples to supplement their data set, regardless of whether the Ministry has used a fifth set.

502-F.06 Correlation

(a) Using the Ministry hammer's best-fit curve established in step Appendix 502-F.04 (or the formula for that curve) determine the expected density at NMin blows.

(b) Subtract the expected density above from the average Ministry manual hammer density determined in Appendix 502-F.04(a) to determine the Ministry mechanical Marshall hammer correlation ("CMin").

(c) Using the Contractor's best fit curve and NCon and the average Ministry manual hammer density determined in Appendix 502-F.04(a) determine the Contractor's mechanical Marshall hammer correlation ("C_{Con}").

502-F.07 Subsequent Testing and Reporting

(a) All subsequent testing by the Ministry and the Contractor shall be conducted on their respective mechanical hammers, delivering the number of blows determined for their respective Marshall Hammer Correlations (NMin or NCon).

(b) Density results determined by the mechanical hammers shall be adjusted by adding, respectively, CMin or CCon and those correlated results reported and used in all applications related to asphalt mix density.

Note: The Ministry has developed an Excel 2016 spreadsheet to perform the calculations and plot the results of this correlation procedure. The spreadsheet is available from the Ministry Representative.

SECTION 503

BASE PREPARATION AND BASE PROCESSING

503.01 Base Preparation – The work shall consist of shaping and compacting the existing granular base course to the designated grade and cross section.

The existing base course material shall be watered, graded and compacted to provide a non-segregated, tightly-knit surface which conforms to the line, grade and cross sections shown on the Drawings or staked by the Ministry Representative to an accuracy of ± 10 mm.

If necessary, the existing surface shall be scarified and bladed to achieve this. No imported materials are to be incorporated into the work, except where failed areas have to be backfilled.

503.01.01 Failed Areas – Where a failed area is present in the existing base, the Contractor shall excavate and dispose of the material, all as directed by the Ministry Representative. The area shall be backfilled using crushed granular base or surfacing aggregate or road-mix as directed by the Ministry Representative and compacted to the same density as the rest of the base course.

503.01.02 Compaction – If watering is required to attain compaction or for any other reason, it shall be applied uniformly from a distributor of the pressure type, equipped with a spray bar mounting nozzles similar to those used on asphalt distributors. Splash plate type distributors or those equipped with spray bars that eject fine streams of water will not be permitted. The distributor must be provided with a satisfactory means for accurately measuring the quantity of water sprayed. If the Ministry Representative so requires, the measuring equipment shall be calibrated under the Ministry Representative's inspection.

The final surface of the base course shall be compacted to a minimum 100% of the Standard Proctor Density obtained by ASTM D698.

503.01.03 Surplus Aggregate – All surplus loose aggregate arising from the base preparation shall be bladed clear into windrows on the shoulder for later incorporation into the shouldering aggregate or bladed to the side slope and trimmed, as directed by the Ministry Representative.

503.02 Base Processing – Where, in the opinion of the Ministry Representative, the base is so deformed, rutted or unsuitable, that it cannot be adequately prepared for priming under the provisions for base preparation, the Ministry Representative shall order in writing how it shall be processed. This work shall be carried out on an Order for Extra Work basis.

503.03 Payment

503.03.01 Base Preparation – Payment for Base Preparation will be at the Unit Prices bid in the Schedule 7 – Schedule of Approximate Quantities and Unit Prices per square metre for the area actually prepared and shall be full compensation for scarifying done to a maximum depth of 40 mm, watering, blading, shaping, trimming, compacting, finishing and maintaining the granular base to the required line, grade, cross section and density.

503.03.02 Repair of Failed Areas – Payment for supply of material used for backfilling and compaction of failed areas will be at the Unit Price bid in Schedule 7 for the material used. Payment for excavating and disposing of the material from the failed areas will be at the Unit Price bid in Schedule 7 for Type D or, where there is no bid price, in accordance with GC 38.00 Change to Work.

503.03.03 No Payment for Incidental Works – Payment will not be made for superintendence, provision or placement of barriers, lights or other equipment which should normally be provided and available for the general performance of the contract.

SECTION 504
PAVEMENT DRAINAGE

DESCRIPTION

504.01 Scope – This Section covers the provision of pavement drainage with the construction of necessary asphalt curbing and drainage outlets using cast iron catch basins or asphalt spillways together with pipe or paved outfalls. Concrete curbs, concrete and corrugated steel catch basins, cast iron grates and frames where applicable are specified by SS 582.

MATERIALS

504.11 Materials – The Contractor shall supply all necessary materials required to construct the Works, conforming to the specifications indicated in SS 320, SS 502, this SS 504, SS 582, and all other provisions of the Contract.

CONSTRUCTION

504.31 Asphalt Curbs

504.31.01 Integral Asphalt Curb – Where specified in the Contract Documents, asphalt concrete curb integral with the asphalt concrete highway pavement shall be constructed to the dimensions indicated on SS Drawing SP504-01.

504.31.02 Machine-Laid Asphalt Curb – Where specified, machine laid (extruded) asphalt concrete curbs shall be constructed in accordance with SS Drawing SP504-01 and as follows:

(a) The type of curb to be placed (Type "A" or Type "B") will be as specified or as determined by the Ministry Representative.

(b) Prior to construction of the curb, the affected pavement area shall be dry and cleaned of loose or deleterious materials. A tack coat of liquid or emulsified asphalt determined by the Ministry Representative or specified in the Special Provisions or Purchase Order shall be uniformly applied at a rate of 0.25 L/m^2 to 0.5 L/m^2, and to a width 0.1 m wider than the proposed curb.

(c) The production and transportation of asphalt mix for curb construction shall be in accordance with SS 502 and SS 952 and shall meet the requirements for Class 1 Fine Mix.

(d) The machine for constructing the curb shall be self-propelled extrusion-type equipment with a material hopper, distributing screw and adjustable curb forming devices capable of placing and compacting the asphalt mix to the required alignment, grade and cross section in an even homogeneous manner free of surface defects. The asphalt concrete mix shall be handled to provide a curb meeting the applicable density requirements of SS 502 and the curbing equipment shall be operated in a manner that ensures a smooth, uniform finished curb alignment. Where hand-forming of curb is required, the finished curb shall conform to all dimensions, alignment, density and finish requirements for machine-laid curb.

504.32 Location of Drainage Outlets – On completion of pavement and curb construction and in conjunction with the installation of any precast concrete barriers, the Contractor shall install or construct drainage outlets at the low point of curves in the vertical alignment and at such other locations designated by the Ministry Representative.

The exact low point on the pavement should be determined by allowing water to flow along the surface to the actual lowest point on the pavement adjacent to the curb or barrier. This low point shall be clearly marked on the pavement.

504.33 Excavation – At the locations designated by the Ministry Representative, the asphalt pavement and any curbing shall be accurately cut away and the highway shoulder excavated to the dimensions consistent with the installation of Cast Iron Catch Basins (SS Drawing SP504-03) or the construction of Asphalt Spillways (SS Drawings SP504-02, SP504-04, and SP504-05). The excavation shall, as nearly as possible, have vertical sides and shall be so done as to cause minimum disturbance of the adjacent compacted base and pavement structures. The outside dimensions of the excavation shall be the smallest compatible with the efficient construction of the drainage facility, as any excess or irregular excavation will be required to be repaired with asphalt mix.

Where applicable, appropriate trench excavation shall be performed to permit the installation and embedment of outlet pipe and associated fittings. Excavation for installation and embedment of outlet pipe shall be considered incidental to the rate bid for pipe installation.

504.34 Drainage Outlets

504.34.01 Cast Iron Catch Basins – Bedding and surrounding mortar for catch basins shall be composed of one part Portland cement to three parts clean well-graded fine aggregate mixed to proper consistency.

Bedding shall be a minimum of 125 mm thick to permit accurate setting of the catch basin in mortar at the correct level true to alignment with curbing, barrier or pavement edge as shown on SS Drawing SP504-03. After securely fixing the adaptor, watertight gasket and connecting pipe, the catch basin shall be surrounded with mortar to the full perimeter of the excavation, and the mortar well tamped and

worked around the connecting pipe for 300 mm from the side of the catch basin.

The mortar shall be allowed to set prior to backfilling and further work.

504.34.02 Asphalt Spillways – Asphalt spillways shall be constructed to the general dimensions shown on SS Drawings SP504-02, SP504-04, and SP504-05, with the necessary hand-forming in accordance with SS 504.31.02(d).

504.34.03 Pipe Outfalls General – Except where paved outfalls are specified, corrugated galvanized metal outlet pipes of the required diameter shall be installed, as shown generally on SS Drawing SP504-03. Should the fill material preclude pipe embedment, the pipe shall be secured to the slope by dual angle iron posts placed at 3 m intervals, as shown. The pipe shall be similarly secured at the bend at the base of the slope and at a point approximately 150 mm from its outlet.

The pipe trench shall be backfilled with excavated material as may be directed by the Ministry Representative and the fill thoroughly compacted.

Class 10 riprap, per SS 205, shall be placed at the pipe outlet as shown on SS Drawing SP504-03.

504.35 Dishing, Spillways and Completion – The asphalt pavement adjacent to the drainage outlet shall be constructed to the profile and cross sections indicated for the type of outlet being installed. The asphalt pavement surface shall be so shaped as to channel all collected water to the outlet orifice and no ridges or other obstructions shall be left which may impede the flow of water to the drainage outlet.

The asphalt pavement mix shall be compacted to the density specified by the Ministry Representative to provide an integral and impervious drainage area. Where the asphalt pavement abuts a cast iron catch basin box, the contact face of the box shall be painted with liquid or emulsified asphalt as specified in SS 502.22.03 prior to placing the asphalt concrete.

On completion, the area shall be cleared and cleaned up to the satisfaction of the Ministry Representative.

MEASUREMENT

504.81 Pavement Drainage Outlets – Pavement drainage outlets will be measured per EACH of the type specified.

504.82 Corrugated Metal Pipe – Corrugated metal pipe will be measured by the METRE along its centreline as installed.

504.83 Asphalt Pavement – Unless otherwise specified, asphalt pavement mix used to form drainage aprons, curbs, spillways and outfalls will be measured by the TONNE.

504.84 Asphalt Curb – Asphalt curb will be measured by the METRE.

PAYMENT

504.91 Pavement Drainage Outlets – Payment for the installation of PAVEMENT DRAINAGE OUTLETS will be at the Contract Unit Price for EACH complete outlet of the type specified. The Contract Unit Price shall be full compensation for supply of cast iron basins, boxes and grates; excavation (see SS 504.33); provision of all materials required for setting basins to correct grade including cement mortar; backfill and compaction; spreading, shaping and compaction of asphalt pavement aprons and spillways; provision and setting of riprap at the outfall and all labour, materials, tools, warning signs, barriers and superintendence required to complete the installation of the drainage outlet.

Payment for Asphalt Pavement mix used will in in accordance with SS 504.93

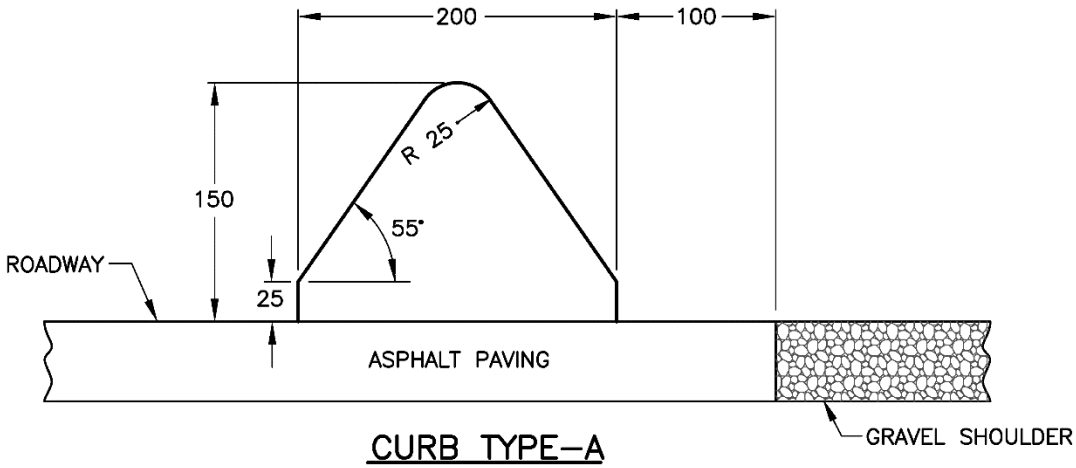
504.92 Corrugated Metal Pipe – Payment for CORRUGATED METAL PIPE will be at the Contract Unit Price per metre. The Contract Unit Price shall be accepted as full compensation for supply and installation including all necessary transport of components, excavation and backfilling; provision of posts, brackets, hoops and hardware; labour for connecting and securing pipe fittings, compaction, superintendence, lights, warning signs, barriers and all other things or services necessary for the proper installation of the pipe.

504.93 Asphalt Pavement – Unless otherwise specified, payment for ASPHALT PAVEMENT used to form drainage aprons, curbs, spillways and outfalls will be at the Contract Unit Price per tonne.

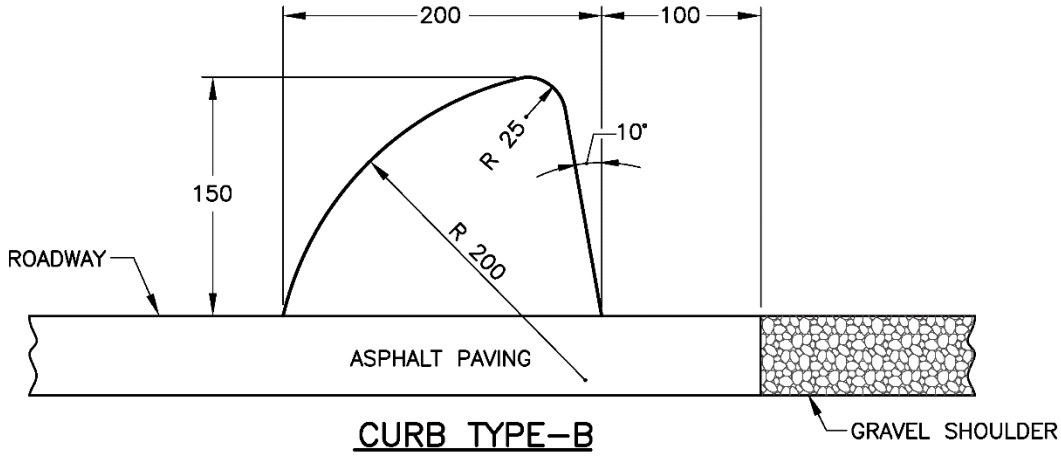
504.94 Asphalt Curb – Payment for ASPHALT CURB will be at the Contract Unit Price per metre of the appropriate curb type placed. The Contract Unit Price shall include the formation of offset curb alignment at drainage outlets in accordance with SS 504.31.

DRAINAGE CURBS TYPE–A, TYPE–B & INTEGRAL ASHPALT

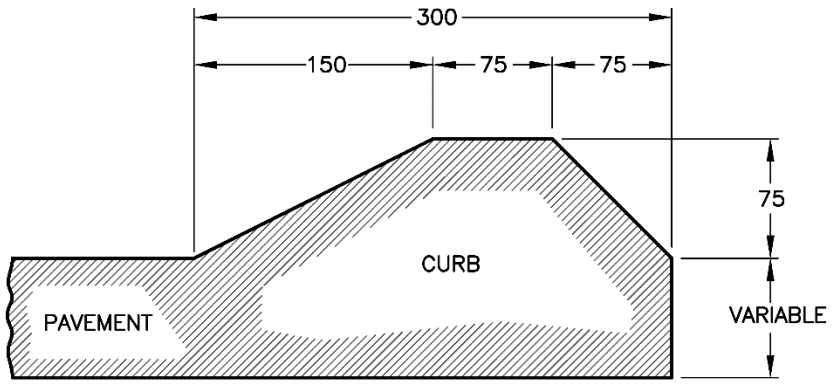
SP504–01



CURB TYPE–A



CURB TYPE–B



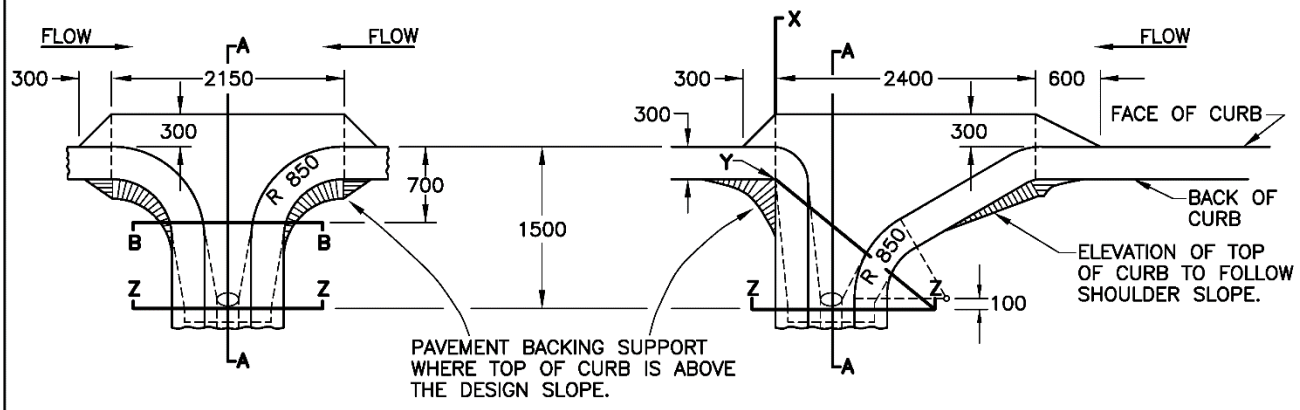
INTEGRAL ASPHALT CURB

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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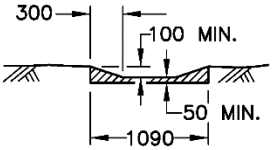
INSTALLATION OF ASPHALT SPILLWAYS

SP504–02



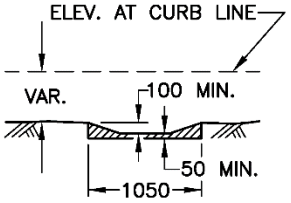
PLAN AT SAG

PLAN ON GRADE



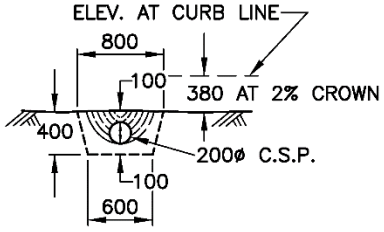
SECTION B–B

COMMON TO PAVED OR PIPED OUTFALL AT "SAG" SPILLWAY



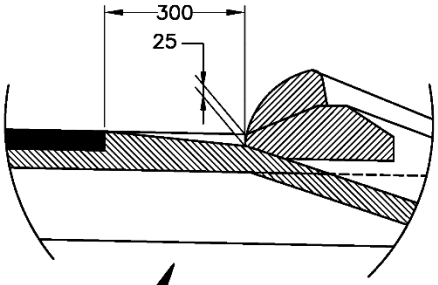
SECTION Z–Z

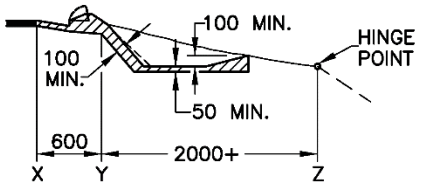
FOR PAVED OUTFALL



SECTION Z–Z

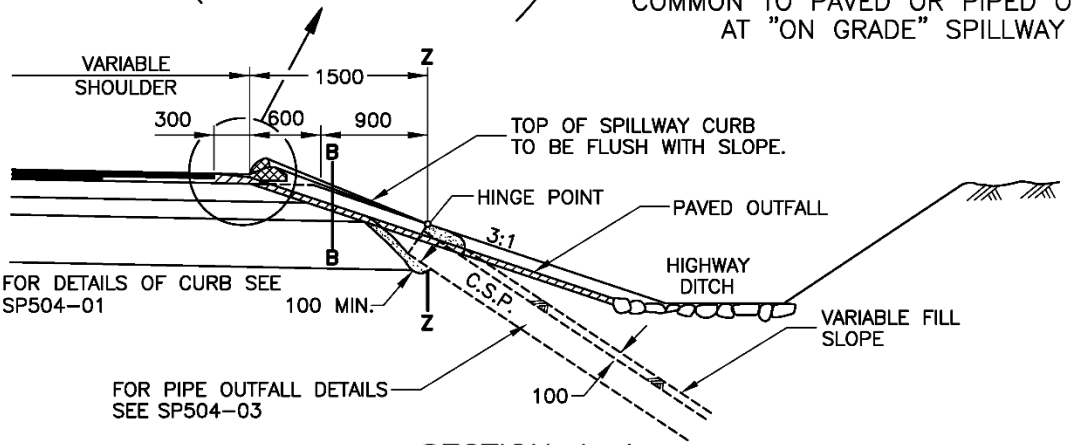
FOR PIPE OUTFALL





SECTION X–Y–Z

COMMON TO PAVED OR PIPED OUTFALL AT "ON GRADE" SPILLWAY



SECTION A–A

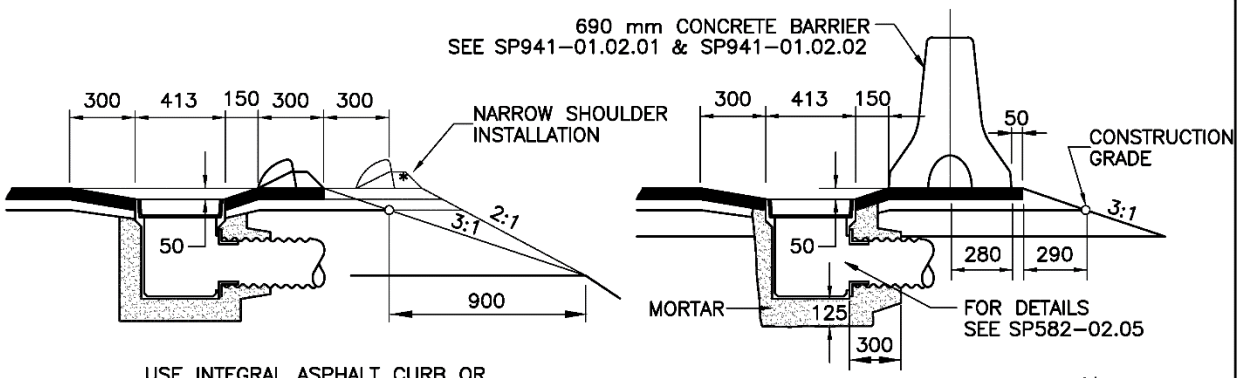
NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

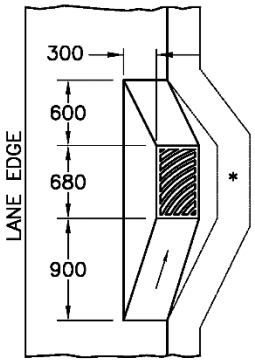
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INSTALLATION OF CURB, BARRIER AND CAST IRON CATCH BASIN

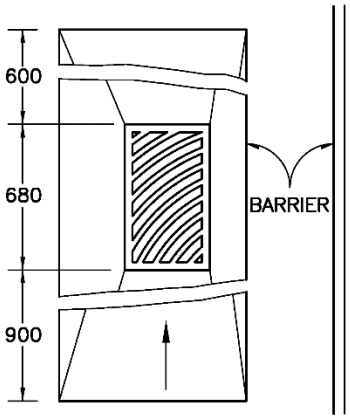
SP504–03



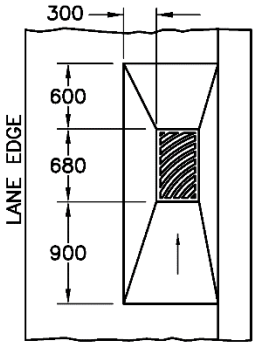
USE INTEGRAL ASPHALT CURB OR TYPE–B CURB (SEE SP504–01)



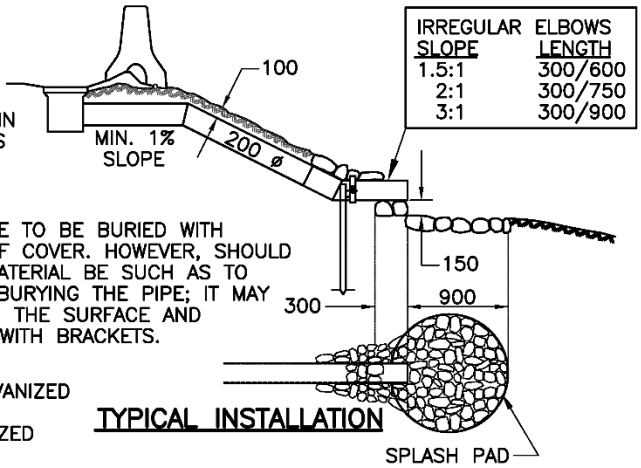
CATCH BASIN AND CURB INSTALLATION FOR NARROW SHOULDERS



CATCH BASIN AT BARRIER

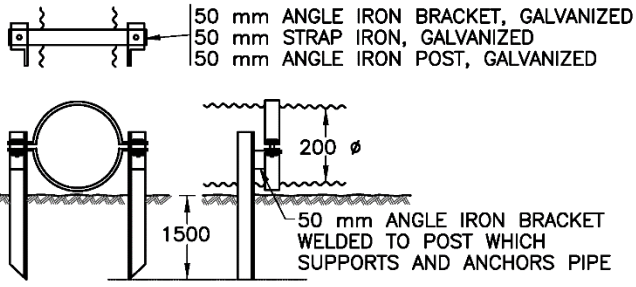


PLACEMENT OF CATCH BASIN WHERE SHOULDER WIDTH IS 1.9 m OR GREATER

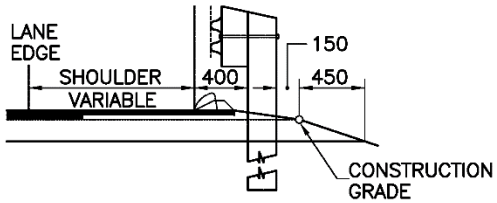


OUTLET PIPE TO BE BURIED WITH 100 mm OF COVER. HOWEVER, SHOULD THE FILL MATERIAL BE SUCH AS TO PRECLUDE BURYING THE PIPE; IT MAY BE LAID ON THE SURFACE AND ANCHORED WITH BRACKETS.

TYPICAL INSTALLATION



DETAIL OF PIPE, ANCHOR BRACKETS AND POSTS



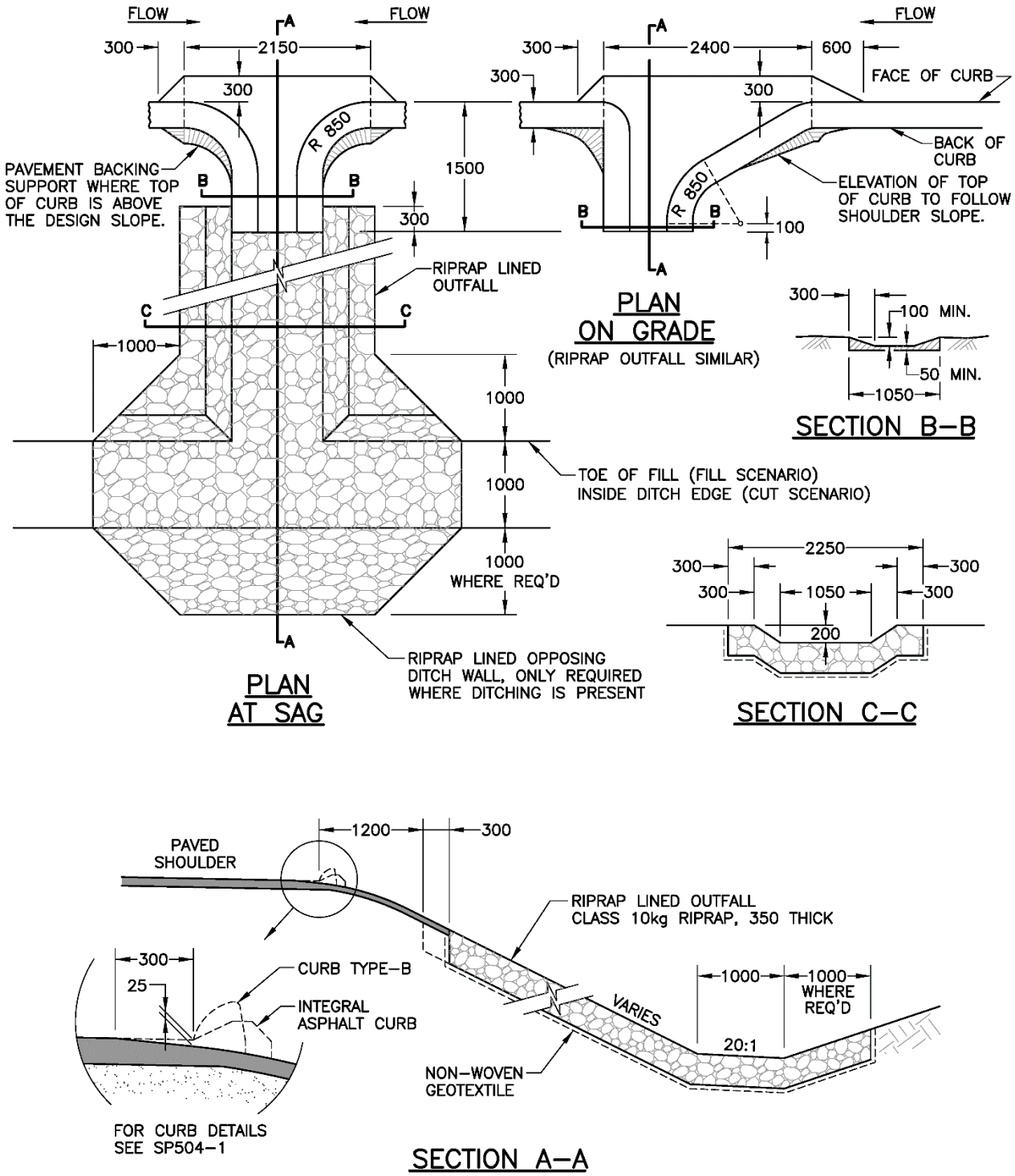
POST AND BEAM GUIDE RAIL

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

INSTALLATION OF ASPHALT SPILLWAYS WITH RIPRAP OUTFALL

SP504–04

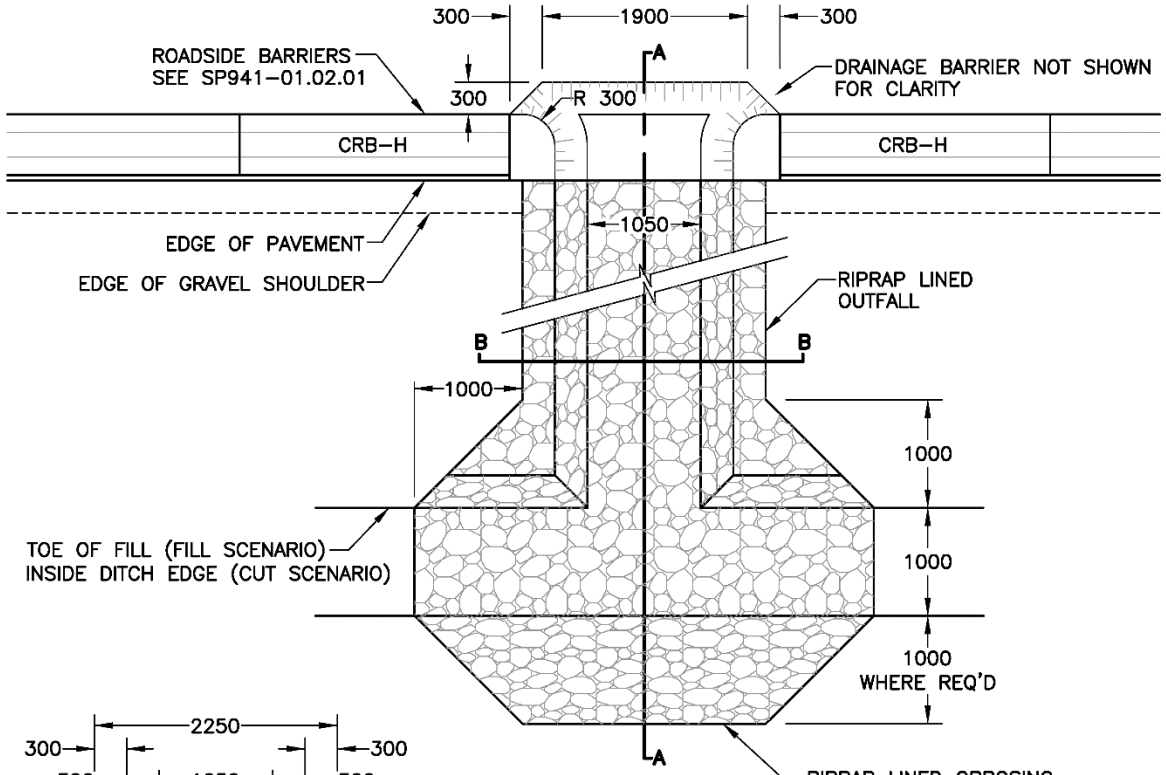


NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

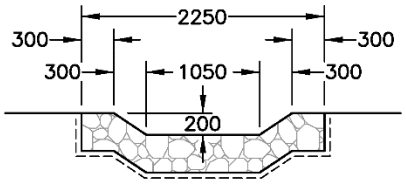
BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

INSTALLATION OF DRAINAGE BARRIER SPILLWAYS WITH RIPRAP OUTFALL

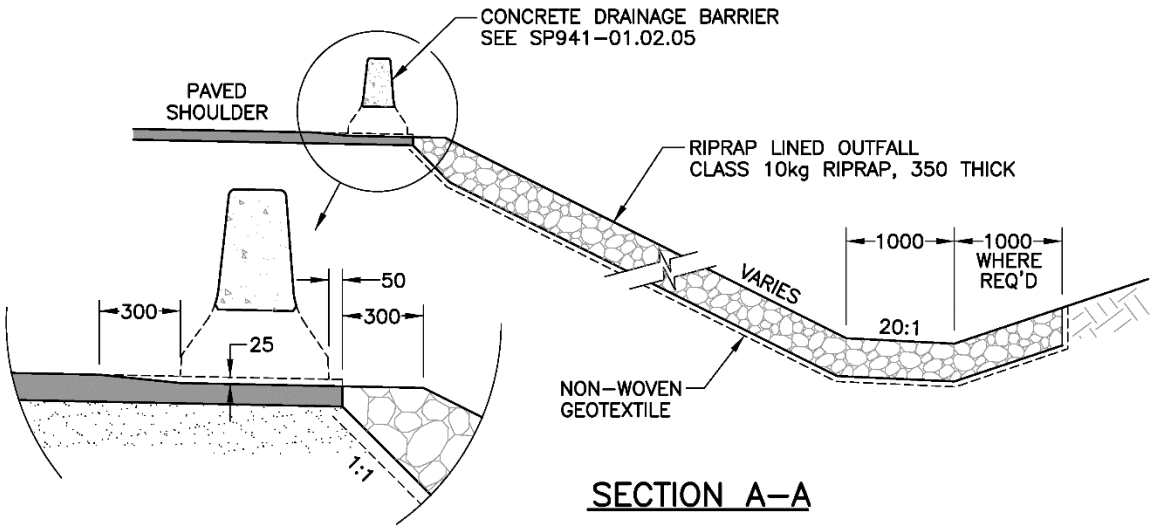
SP504–05



PLAN AT SAG



SECTION B–B



SECTION A–A

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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SECTION 505

USE OF RECLAIMED ASPHALT PAVEMENT IN ASPHALT PAVEMENT CONSTRUCTION

505.01 Preliminary and General – The work covered by this section covers the use of Reclaimed Asphalt Pavement (RAP) material in hot mix asphalt construction. This specification allows for the use of both Classified and Unclassified RAP.

The Contractor's development and processing of RAP into an end paving product must meet SS 502 Asphalt Pavement Construction (EPS) for Highway Construction.

The maximum allowable percentage of RAP allowed in asphalt pavement will be based on the classification of the roadway and the location of the mix in the pavement structure or as identified in the Special Provisions.

505.02 Roadway Categories – Roadways are divided into the following two categories:

(a) Category A – All numbered Highways.

(b) Category B – Other Roads.

The roadway Category shall Category A, unless otherwise identified in the Special Provisions.

505.03 Definitions

(a) Reclaimed Asphalt Pavement (RAP) – Removed and/or reprocessed pavement materials containing asphalt and aggregates. These materials are generated when asphalt pavements are removed for reconstruction and/or resurfacing. When properly screened and crushed, it consists of high-quality aggregate coated by asphalt cement.

(b) Classified RAP – RAP obtained from Ministry roadways.

(c) Unclassified RAP – RAP obtained from non-Ministry sources or mixed with RAP from other sources.

(d) Rheology – The study of deformation and flow of matter. For asphalt binder rheology, it refers to the flow characteristics of the asphalt binder in the asphalt mix. Deformation and flow of the asphalt binder in asphalt mix is important in determining asphalt pavement performance.

505.04 Allowable RAP Addition Rates – The maximum RAP allowed in the asphalt mix shall be determined by the contribution of the RAP Asphalt Cement (AC) towards the total AC content in the mix by weight as per the percentages specified in Table 505-A.

The Contractor shall notify the Ministry Representative, a minimum of fourteen (14) days in advance of paving if RAP is proposed to be used in the project, and shall fill all fields as applicable and sign a copy of the "RAP Record Sheet" provided in APPENDIX 505-B and submit it to the Ministry Representative a minimum of ten (10) days in advance of the proposed start date for paving, unless the project Special Provisions state differently.

The amount of total AC replaced by AC in the RAP will be calculated as follows:

$$\%\ \boldsymbol{AC\ Replacement} = \frac{(\boldsymbol{a\ x\ b})}{\boldsymbol{c}}$$

Where;

a = AC content of RAP

b = RAP percent in mixture by total weight of mix

c = Total Percent AC content in mixture

Inclusion of RAP into the hot mix asphalt will be as per the maximum allowable percentage of AC replacement unless otherwise noted in the Special Provisions.

Table 505-A: Maximum Percent RAP AC Replacement Allowed in Asphalt Mix

Road Classification	Top Lift	Lower Lifts
Category A	15%	30%
Category B	30%	30%

505.05 Materials

505.05.01 RAP – The Contractor shall fulfill or exceed the requirements of APPENDIX 505-A for the management of RAP materials from the time of collection through processing, mix design and quality control practices during the production of asphalt mixtures containing RAP.

The Contractor shall provide documentation to the Ministry Representative that best practices have been followed in the handling, classification, and storage of RAP material, prior to being allowed to use the RAP.

505.05.02 Asphalt Cement – When the Percent RAP AC Replacement amount is greater than 15%, the blended AC must meet the penetration and viscosity requirements of the specified asphalt cement specified in the Special Provisions (when penetration graded asphalt cement is specified) or the temperature requirements of the specified Performance Graded Asphalt Cement (PGAC) when a PGAC is specified or substituted for penetration graded asphalt cement.

505.05.03 Use of Rejuvenators – Usage of Softening agents, rejuvenators or recycling agents will not be permitted.

505.05.04 Use of Recycled Asphalt Shingles – Usage of recycled asphalt shingles will not be permitted.

505.06 RAP Sampling – Sampling of the RAP shall be as per APPENDIX 505-A.

505.07 Quality Control – The quality control testing requirements for RAP shall be based on the percent of AC replacement in the asphalt mix and comply with the requirements listed in Table 505-B and Table 505-C, and the corresponding tests specified in SS 505.07.01.

Table 505-B: Quality Control Testing Requirements

RAP Category	% AC Replacement	Required Tests
Classified RAP	≤ 15%	• Basic Tests
	> 15%	• Basic Tests • Consensus Tests • AC Rheology Testing
Unclassified RAP	≤ 15%	• Basic Tests • Consensus Tests
	> 15%	• Basic Tests • Consensus Tests • AC Rheology Testing

Table 505-C: Minimum Quality Control Test Frequencies

Test	Minimum Frequency
Asphalt Content	One per 750 tonnes
Gradation	One per 750 tonnes
Percent Fracture (%)	One per 750 tonnes
Specific Gravity of coarse fraction of RAP	Minimum of one per 3000 tonnes or three per stockpile
Specific Gravity of fine fraction of RAP	Minimum of one per 3000 tonnes or three per stockpile
Maximum Micro Deval Abrasion loss factor (%)	Minimum of one per 3000 tonnes or three per stockpile
Fine aggregate angularity	Minimum of one per 3000 tonnes or three per stockpile
Flat and elongated particles (For Superpave Only)	Minimum of one per 3000 tonnes or three per stockpile
AC Rheology	Minimum of one per 3000 tonnes of RAP or a minimum of three tests per project.

505.07.01 RAP Aggregate Testing

(a) Basic Tests – Individual representative RAP samples shall be tested to determine moisture content, gradation, percent fracture and asphalt content.

Tests to determine the specific gravity of the coarse and fine fractions of RAP shall be completed on the combined aggregates (obtained by combining the individual reclaimed aggregates retained after the extraction of the asphalt cement).

The following basic properties shall be determined for the proposed RAP:

(i) Moisture Content;

(ii) Asphalt Content;

(iii) Gradation;

(iv) Percent Fracture; and

(v) Specific Gravity of coarse and fine fractions (for mix design purposes).

(b) Consensus Tests – The tests for the consensus properties shall be completed on the combined aggregates (virgin aggregates and reclaimed aggregates) mixed in the proportions proposed for the mix design.

The following aggregate consensus properties shall be determined for the combined aggregate samples:

(i) Maximum Micro Deval Abrasion loss factor (%);

(ii) Standard Test Methods for un-compacted void content of fine aggregate (as influenced by particle shape, surface texture and grading) (ASTM C1252); and

(iii) Percentage of flat and elongated particles (for Superpave mixes only).

505.07.02 AC Rheology Testing – When AC rheology testing and design is required, the blended AC must meet the penetration and viscosity requirements of the specified AC type.

For penetration graded asphalt, rheology is measured in terms of penetration at 25°C and viscosity at 60˚C.

For performance grade asphalt cements (PGAC), rheology is tested as per the procedures outlined in AASHTO R 29 *Standard Practice for Grading or Verifying the Performance Grade (PG) of an Asphalt Binder* and reported in terms of high and low temperature service ratings, i.e. PG 58-34.

Since the rheological properties of asphalt binder vary with temperature, rheological characterization involves two key considerations:

(a) To compare different asphalt binders, their rheological properties must be measured at some common reference temperature.

(b) To fully characterize an asphalt binder, its rheological properties must be examined over the range of temperatures that it may encounter during its life.

For mixes specified to use PGAC, the blended AC must meet the temperature requirements of the specified PGAC.

For asphalt mixtures containing RAP and specified to use Performance Graded (PG) asphalts, the RAP rheology and the grade of virgin asphalt to be used shall be determined according to AASHTO R 35 *Standard Practice for Superpave Volumetric Design for Asphalt Mixtures*.

505.08 Asphalt Mix Design – RAP shall be included at the proposed addition rates during the mix design. The Contractor shall submit a mix design to the Ministry Representative for review at least seven (7) days prior to the start of asphalt mix production.

505.08.01 Additional Mix Design Information – In addition to the information required in SS 502.08.04 Asphalt Mix Design Submittals, the following information shall be provided in the mix design submittal for the asphalt mixes containing RAP:

(a) All RAP aggregate gradations for each RAP product;

(b) Asphalt Content for RAP;

(c) Results for Consensus properties of combined aggregates as specified in Section 505.07.01 (when applicable);

(d) Design RAP Rheology test results; and

(e) Blending Charts for the virgin and reclaimed AC. (when applicable)

505.08.02 Changes in RAP – Should a change in the source or addition rate of RAP be made after the mix design has been approved, a new mix design and/or job mix formula shall be submitted to the Ministry Representative for review.

505.09 Payment Adjustments – SS 502 – PAYMENT ADJUSTMENTS (SS 502.50 through SS 502.58 inclusive) shall apply to all work utilizing reclaimed asphalt pavement in the asphalt mix unless otherwise specified in the Contract.

APPENDIX 505-A

RAP MANAGEMENT BEST PRACTICES

505-A.01 Best Practices – A best practice in the context of this Standard Specification is a method or technique that has consistently shown results superior to those achieved with other means, and that is used as a benchmark. A "best" practice may evolve to become better as improvements are discovered. Best practice is considered to describe the process of developing and following a standard way of doing things that are commonly achieved in the industry.

505-A.02 RAP Sources – RAP may be obtained from several sources. The common sources of RAP are:

(a) **Cold Milling –** This is the most common source of RAP. The milling process should be closely examined to make sure that the milled material is not contaminated with soil, base materials, paving geotextiles or other foreign material (sulphur, asbestos, rubber etc.). The milled material that becomes contaminated should be stockpiled separately from RAP to be used in asphalt mix.

A special milling operation may also be beneficial when it is desirable to mill the surface layer in one pass and the underlying layers in a second pass because the surface course millings may contain aggregates with higher fractured aggregates that could be incorporated in the new surface layers.

(b) **Full Depth Pavement Removal –** RAP can also be obtained from the removal of the existing pavement using a bulldozer or a backhoe. This process typically results in large chunks of pavement that may be contaminated with underlying soils. This contaminated material should be stored in a separate stockpile and not to be used in the hot mix asphalt.

(c) **Asphalt Plant Waste –** This includes the waste generated during plant start-up, transition between mixes, plant clean out, mix rejected from projects and excess mix produced that could not be placed.

This material usually has fewer fines than the typical RAP since it wasn't milled or broken up during pavement removal and the AC is less aged than RAP since it hasn't been subjected to environmental aging.

This material should be stockpiled and tested separately from the other RAP.

505-A.03 RAP Contamination – Best practice for RAP stockpiles is to keep them free from contaminants. RAP stockpiles should be treated as a valuable material. Truck drivers bringing the material on the site should ensure that unwanted debris or contaminated material does not end up in the RAP stockpile.

The plant QC personnel and the loader operators should continuously monitor processed and unprocessed RAP stockpiles to make sure that they do not contain deleterious materials. If any contaminants are found, they should be removed immediately so that they are not covered up with other RAP brought on to the yard.

505-A.04 RAP Categories – RAP is categorized as:

(a) Classified RAP: RAP obtained from Ministry sources.

(b) Unclassified RAP : RAP obtained from non-Ministry sources or mixed with RAP from other projects.

505-A.05 RAP Processing – Best practice for RAP processing involves one or more steps to create consistent materials. Screening is used to separate sizes. In some cases, it may be desirable to screen or fractionate RAP to coarse and fine fractions. RAP separation based on sizes increases quality and reduces variability in the RAP properties.

For stockpiles of RAP from multiple sources, particularly stockpiles containing oversize fragments of RAP or pavement slabs, the material should be processed to produce RAP with a maximum size of 37.5 mm for use in hot mix asphalt.

Further processing of RAP may include both screening and crushing to produce a uniform gradation, AC content and other properties. Since crushing RAP will create more aggregate fines, it is best to set up the crushing operation so that the RAP is screened before it enters the crusher.

RAP processing shall occur as early as possible in the construction planning process.

505-A.06 RAP Stockpiling – The best practice to minimize the accumulation of moisture in stockpiles is to cover the stockpile with a shelter, tarp or building to prevent precipitation from getting to the RAP.

RAP stockpiles should be placed on a base with adequate drainage and constructed with minimal segregation. Arc-shaped, conical, uniform stockpiles are preferred for storing processed or unprocessed RAP. No contamination of the RAP from the stockpile base materials should occur.

Use of heavy equipment on top of the RAP stockpile should be minimized to avoid compaction of the RAP. It is also recommended that the RAP stockpiles be limited to 4 m in height to reduce the potential for self-consolidation of the stockpile.

505-A.07 RAP Sampling – Best practice is for representative RAP samples to be collected from the stockpile prepared for the project. At least one sample per

750 tonnes of RAP in the stockpile or a minimum of ten samples per stockpile should be taken and retained for testing.

If the asphalt mix from an existing pavement is to be used as RAP, 150 mm diameter cores should be extracted at a frequency of at least one core every 1.5 kilometre in each lane prior to the start of the project. The asphalt mix from the pavement lift to be recycled should be tested to determine the properties of the aggregates and the asphalt cement in the pavement.

505-A.08 Sampling Method – Best practice for sampling aggregates applies to the sampling of RAP as well. RAP stockpiles should be sampled as they are being built at the location where they will be fed into the asphalt plant. Sampling at the time the stockpile is built is the best practice and will be easier and more representative of the stockpile compared to samples taken later after the formation of the crust on the face of the RAP stockpile.

ASTM D75 or AASHTO R 90 sampling procedures, normally used for virgin aggregates, should also be used to sample RAP aggregates.

505-A.09 Minimum Test Frequencies – The quality control tests on the RAP and the extracted aggregates should be completed at the minimum frequencies specified in Table 505-C.

All RAP aggregates retained after extraction should be combined together into one sample. The combined sample should then be sieved and split into coarse and fine fractions and used to determine the specific gravity of the recycled aggregates.

505-A.10 Stockpile Management – When a stockpile reaches the desired quantity and has been sampled and characterized, no additional RAP material should be added to it. Subsequent RAP material should be stockpiled in a separate stockpile and characterized in the same manner. This process should continue such that characterized stockpiles are not compromised by new RAP materials.

Samples from the RAP stockpile should be taken and the testing completed as per the minimum test frequencies specified in Table 505-C.

APPENDIX 505-B

RAP RECORD SHEET

(See SS 505.04)

Contractor: _____

Project Name per Signed Contract: _____

Project Number: _____

Numbered Highway Name: (If Applicable): _____

Road Name: (If Applicable): _____

Segment No.	Tonnage	RAP %	Classified RAP	Unclassified RAP	Top Lift %	Bottom Lift %	LKI		
							Left/Right	Start	Finish

Filled by (Name and Date): _____

Address and Contacts: _____

Submitted by: _____

Name: _____

Signature and Date: _____

SECTION 507
PRODUCTION AND STOCKPILING OF ASPHALT MIX

GENERAL

507.01 General – This Section covers the supply of aggregate, mixing it with a specified asphalt material in a central plant and placing the produced mix in a stockpile.

MATERIALS

507.10 Aggregate

507.10.01 Gradation Limits – The Contractor shall supply aggregate with gradation wholly within the limits listed in Table 507-A when tested in accordance with ASTM C136 and ASTM C117 (wash test).

Once a crushed aggregate gradation has been accepted, the maximum permissible variation of the mean of any four consecutive tests from the accepted gradation curve shall be within the limits specified in Table 507-B.

Should there be a substantial change in the type of aggregate exposed as the work proceeds, the Ministry Representative may authorize a change in the mean gradation.

507.10.02 Fractured Faces – A minimum of 85% by count of all aggregate particles retained on the 4.75 mm sieve shall have at least two fractured faces or shall be naturally angular with sharp edges (ASTM D5821). If this requirement cannot be met for a specified crushed aggregate type then the Contractor shall, at no additional charge, crush aggregate to the next smaller size or such intermediate size as the Ministry Representative may direct, to meet the requirements.

507.10.03 Objectionable Matter – All aggregates shall be free from coatings of clay, silt or other objectionable matter and shall contain no clay lumps or other aggregations of fine-grained materials.

507.10.04 Physical Properties – Crushed aggregate shall conform to the following requirements:

(a) The sand equivalent value in accordance with ASTM D2419 shall not be less than 40.

(b) The degradation factor in accordance with ASTM D6928 shall be not more than 20.

(c) In the absence of satisfactory performance records for a particular source of aggregate, its soundness shall be tested according to ASTM C88. Aggregate so tested shall be considered unsatisfactory if the loss after five cycles exceeds 20% for coarse aggregate or 25% for fine aggregate.

(d) Micro Deval maximum abrasion loss shall not be greater than 17% as tested under ASTM D6928.

Table 507-A: Gradation Requirements for Aggregate

Sieve Size (mm)	Percent Passing (By Mass)	
	Patch Mix	Paving Mix
19.0	---	100
16.0	---	100 – 92
13.2	---	100 – 80
9.5	100	90 – 65
4.75	85 – 55	75 – 45
2.36	65 – 40	60 – 30
1.18	50 – 25	45 – 20
0.300	30 – 15	25 – 10
0.150	18 – 8	17 – 5
0.075	12 – 5	9 – 2

Table 507-B: Maximum Permissible Variation of Crushed Aggregate Gradation from the Gradation Curve

Sieve Size	Variation Limit (% Passing)
4.75 mm and larger	± 5.0
1.18 mm to 4.75 mm	± 3.5
0.300 mm to 1.18 mm	± 2.0
0.075 mm to 0.300 mm	± 1.0

507.10.05 Fine Aggregate – For testing purposes, fine aggregate shall be all mineral matter passing the 4.75 mm sieve including mineral fillers. It shall consist of natural and/or manufactured material derived by crushing stone, slag or gravel.

507.10.06 Mineral Filler – Mineral filler shall:

(a) Consist of all mineral matter passing the 0.425 mm sieve.

(b) Be dry, free from organic matter, clay particles or lumps.

(c) Be non-plastic when tested in accordance with ASTM D4318.

507.10.07 Mineral Dust – Mineral dust shall consist of all mineral matter passing the 0.075 mm sieve.

507.11 Asphalt Material

507.11.01 Type – The type of asphalt material to be used will be specified in the Contract's Special Provisions.

507.11.02 Supply – The Contractor shall supply the Liquid Asphalts or High Float Emulsified Asphalts, as specified in the Contract, to meet the requirements of SS 952.

CONSTRUCTION

507.20 Mix Design – The Contractor will carry out the Asphalt Mix design, in accordance with specifications provided in the Contract and to for approval by the Ministry.

507.20.01 Sample Supply – At least seven days prior to commencement of mixing, the Contractor shall submit the following to the Ministry Representative:

(a) A 50 kg sample of aggregate which is representative of the aggregate to be used on the project.

(b) If available, a 30 kg sample of plant produced asphalt mix whose gradation is deemed to be representative of the gradation of the aggregate to be used on the project.

(c) A 5 L sample of the type of asphalt to be used on the project.

507.20.02 Variation from Design Gradation – Once the design grading curve is established, the maximum permissible variation will be as specified under SS 507.02.01

507.21 Mixing Requirements

507.21.01 Mixing Plant – Mixing operations shall be performed in a mixing plant capable of producing the asphalt mix within the required specifications.

507.21.02 Aggregate Moisture Content – The maximum moisture content of the aggregate immediately prior to mixing with a liquid asphalt (MC or SC) shall not be over 2% by mass of dry aggregate.

507.21.03 Mix Temperatures – Temperatures of the mix shall be generally in accordance with the limits specified in Table 507-C. Mix temperatures may be lowered to reduce the emission of "blue smoke" from the discharge chute.

507.21.04 Asphalt Content

(a) AC Variation Limit – The asphalt content in the mix shall not vary from the design asphalt content by more than 0.3 by mass of total aggregate.

(b) AC Content Increase – The Ministry reserves the right to increase the asphalt content in the mix. The Contractor shall be compensated for the additional quantity of Liquid Asphalts or High Float Emulsified Asphalts used on a Force Account Basis.

507.21.05 Asphalt Mix Moisture Content – The asphalt mix shall not contain more than 1% moisture by mass at the discharge from the plant.

507.21.06 Asphalt Mix Uniformity – The asphalt mix shall be uniform in appearance with all particles thoroughly coated. The mixing time will be increased if the mix is not completely coated with asphalt.

Table 507-C: Mix Temperatures

Type of Asphalt	Asphalt Storage Temperature Range (°C)	Mix Temperature Range (°C)
SC – 250	65 – 80	60 – 80
MC – 250	65 – 80	60 – 80
HF – 500M	55 – 65	100 – 120*
HF – 1000M	55 – 65	100 – 120*
* High temperatures are used when the mix is hauled directly to the road and when dictated by high moisture in the aggregate.		

507.22 Hauling – When the haul distance from the mixing site to the stockpile is greater than 1 km, the hauling vehicles shall be equipped with and use a tarpaulin of such size as to completely protect the asphalt mix from the weather and inhibit premature curing of the mix by aeration.

The tarpaulin shall be double quilted canvas or other approved material treated to prevent absorption of the asphalt.

507.23 Stockpiling

507.23.01 Stockpile Location – The exact location of the stockpile site will be determined by the Ministry. The Contractor shall place the cold mix within the boundaries of the designated stockpile site.

507.23.02 Site Preparation – The Contractor shall prepare the stockpile site, which shall be shaped to a uniform smooth surface and graded to ensure good drainage.

507.23.03 Handling – Temporary stockpiling of the mix on the ground or in storage bins will be allowed, however, the Contractor will be required to minimize the handling of the mix to prevent premature curing of the mix.

507.23.04 Conveyors – If conveyor belts are used for temporary stockpiling or final stockpile construction, the

free fall distance from the conveyor belt to the stockpile shall not exceed 2 m.

507.23.05 Layer Depth – If the stockpile is constructed using a front-end loader or end dumping by truck, the mix shall be deposited in layers of approximately 1 m in depth.

507.23.06 Protection from Contamination and Consolidation – Plank or protected runways shall be provided for operating equipment on stockpiles, when the Ministry Representative deems them necessary, to prevent contaminants being tracked onto the stockpile and to minimize consolidation of the mix.

507.23.07 Stockpile Height – The final stockpile shall be constructed in such a manner that height is not more than 4 m.

507.23.08 Precipitation – Mixing, hauling or stockpiling will not be permitted during periods of precipitation.

MEASUREMENT

507.80 Measurement – The asphalt mix will be measured in TONNES and the asphalt material(s) in LITRES, corrected to 15°C.

PAYMENT

507.90 Payment – Payment for ASPHALT MIX in stockpile will be at the contract Unit Price per tonne.

The unit price will be full compensation for preparing the stockpile site; supplying and heating the aggregate; supplying, heating and storing the asphalt material; mixing, hauling and stockpiling the mix.

SECTION 508
GRADED AGGREGATE SEAL COAT (EPS)

GENERAL

508.01 Preliminary and General – This section describes the materials, equipment, professional standards and the end product requirements for the construction of graded aggregate seal coats. This section shall be read and construed together with all applicable Standard Specifications for Highway Construction. Should this Section conflict with any other Section, this Section shall take precedence unless otherwise stated in the Special Provisions.

508.02 General Description of Work – Surface seals shall consist of the application of a selected asphalt emulsion and embedment of the selected aggregate to provide a new, uniformly textured surface. This surface shall be impervious to moisture, be skid-resistant and have good riding qualities.

508.03 Classes of Graded Aggregate Seals – There are six classes of Graded Aggregate Seal, including but are not limited to graded aggregate seals, a coarse or fine sand seal or any other graded aggregate surface treatment outlined in the Special Provisions.

The Classes and their common applications are:

- **Class A** – Double seals on gravel highways
- **Class B** – Double seals on gravel highways, and single seals low traffic paved roads
- **Class C and Mod C** – Major Highways
- **Class D** – Major Highways
- **Class E** – Residential areas and second seals

508.04 Definitions

(a) Coarse Aggregate – Coarse aggregate is defined as all material retained on or above the 4.75 mm sieve.

(b) Fine Aggregate – Fine aggregate is defined as all materials passing the 4.75 mm sieve.

(c) Mineral Filler and Mineral Dust – Mineral filler shall consist of all mineral matter passing the 0.600 mm sieve and the mineral dust shall consist of all mineral matter passing the 0.075 mm sieve.

(d) Blinding Sand – Fine aggregate used as large aggregate support.

MATERIALS

508.11 Asphalt Materials – The Contractor shall supply the asphalt emulsion as specified in the Special Provisions, which shall meet the requirements of SS 952.

Asphalt(s) to be used on surface seals may be standard rapid curing cutback asphalt; medium curing cutback asphalts; or anionic or cationic emulsions including high float emulsified asphalts. Types and grades of asphalt binder proposed for use on the Project will be specified in the Special Provisions.

The Ministry Representative may require the use of anti-stripping additives or any other materials formulated to improve the coating and adhesive properties of the asphalt binder to the cover aggregate. Instructions concerning the type of additive to be supplied and the method of incorporation with the asphalt binder will be specified in the Special Provisions.

508.12 Aggregates

508.12.01 Mineral Aggregate Supply – The Contractor shall supply mineral cover aggregate meeting the requirements described hereunder unless specifically described otherwise in the Special Provisions.

The Ministry may supply the aggregates. If the Ministry does supply aggregates, then the Ministry will supply quality control information, if available, to the Contractor. The Contractor must verify and be satisfied that the Ministry supplied aggregates will meet the quantity and quality requirements prior to submitting a bid.

508.12.02 Mineral Aggregate: Test Methods – Mineral Aggregate will be sampled and tested in accordance with the standard ASTM procedures and tests listed in Table 508-A.

508.12.03 Test Requirements – Mineral cover aggregate shall consist of clean, sound, hard, durable particles or fragments of sand, gravel or crushed stones or combination thereof, containing a minimum of thin elongated or flaky pieces. It shall be substantially free from loosely bonded aggregations, clay lumps or other objectionable matter and shall not be markedly hydrophilic in character.

The mineral cover aggregate, when tested by the methods listed in Table 508-A, shall meet the following criteria:

(a) Clay Lumps – Coarse aggregate shall not contain more than 0.5% by mass of clay lumps or other aggregation of fine material.

(b) Micro-Deval – No individual test shall have a loss factor of more than 20%.

(c) Flat and Elongated Particles – The percent flat and or elongated particles must be less than 10% using 4:1 ratio.

(d) Fracture – Contrary to ASTM D5821, only those particles retained on or above the 4.75 mm sieve with two or more fractured faces shall be measured. The

minimum % by mass, for coarse aggregate shall be as follows:

(i) Classes A & B 75%

(ii) Classes C, Mod C, & D 85%

(iii) Class E not applicable

Table 508-A: Test Methods for Mineral Aggregate

ASTM Test	Name of Test
C117	Standard Test Method for Materials Finer than 75-µm (No. 200) Sieve in Mineral Aggregates by WashingWash Test of Aggregates.
C127	Bulk Relative Density and Absorption of Coarse Aggregates
C128	Bulk Relative Density and Absorption of Fine Aggregates
C136	Standard Test Method for Sieve Analysis of Fine and Coarse AggregatesSieve Analysis of Aggregates.
C142	Standard Test Method for Clay Lumps and Friable Particles in Aggregates.
C566	Moisture Content of Aggregate by Drying
D75	Standard Practice for Sampling Aggregates.
D4318	Liquid Plastic Limit Determination
D4791	Standard Test Method for Flat Particles, Elongated Particles, or Flat and Elongated Particles in Coarse Aggregate Flat and Elongated Particles.
D5821	Standard Test Method for Determining the Percentage of Fractured Particles in Coarse AggregateFracture Count on Coarse Aggregate.
D6928	Standard Test Method for Resistance of Coarse Aggregate to Degradation by Abrasion in the Micro-Deval ApparatusMicro-Deval.

508.12.04 Graded Aggregate Seal

(a) Aggregate Gradation – Gradations shall meet the requirements stated in Table 508-B for each Class of graded aggregate seal.

(b) Variation Limits – For all Classes of seal except Class E, once the grading curve is established within the above limits, the maximum permissible variation of the mean of any four consecutive tests from the grading curve shall be within the tolerances specified in Table 508-C.

These tolerances do not waive the requirement that the running average of four consecutive tests must be maintained at all times inside the limits specified in Table 508-B. Should the Contractor not be able to maintain the grading within the variation limits specified above they shall be required to construct a second stockpile. This does not alleviate the Contractor from consistently maintaining the grading curve within the tolerances indicated in the gradation curves.

Table 508-B: Gradation Limits for Graded Aggregate Seal

Sieve size (mm)	% Passing by Mass by Class					
	A	B	C	Mod C	D	E
19	100	-	-	-	-	-
16	-	100	100	100	-	-
12.5[1]	60-90	60-90	-	-	100	-
9.5	40-80	35-75	30-70	25 -55	30-80	100
4.75	20-60	15-50	25-45	7 – 30	25-45	0-30
0.600	0-25	0-15	5-20	0 -10	5-20	0-10
0.075	0-7	0-5	0-3	0 – 3	0-3	0-3
Ratio[2] (9.5/4.75)	-	-	-	1 – 1.5	-	-

Notes:

[1] The Contractor may use either the 12.5 mm or the 13.2 mm sieve, at the Contractor's option.

[2] The "Ratio" is the ratio of the mass of material retained on the 9.5 mm sieve, divided by the mass of material retained on the 4.75 mm sieve.

Table 508-C: Variation Limits

Sieve Size (mm)	Maximum Permissible Tolerance % Passing by Mass
4.75 to 19.0	± 5.0
0.600	± 3.5
0.075	± 1.0

508.12.05 Aggregate Characteristics

(a) Coarse Aggregate – Coarse aggregate shall consist of crushed stone or gravel, or a combination thereof; or